

Voltage References

Voltage references

Voltage references are blocks that produce an output voltage that is independent of **PVT** variations:

V: Supply voltage

T: Temperature

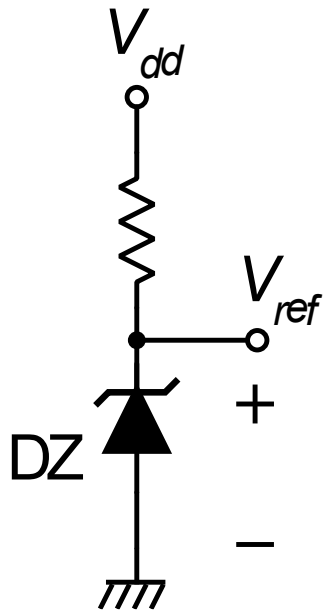
P: Process errors (Matching error)

Voltage references are used for:

- Providing an absolute reference voltage for ADCs and DACs
- Providing an absolute reference voltage for stimulating sensors or other external devices that require precise control voltages and/or currents.
- Creating **constant bias voltages and currents** when required

Possible reference voltage sources

- Zener Diodes



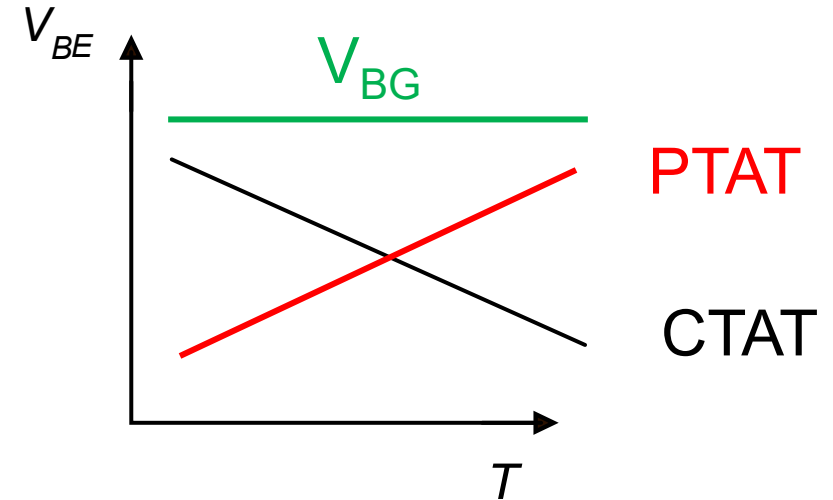
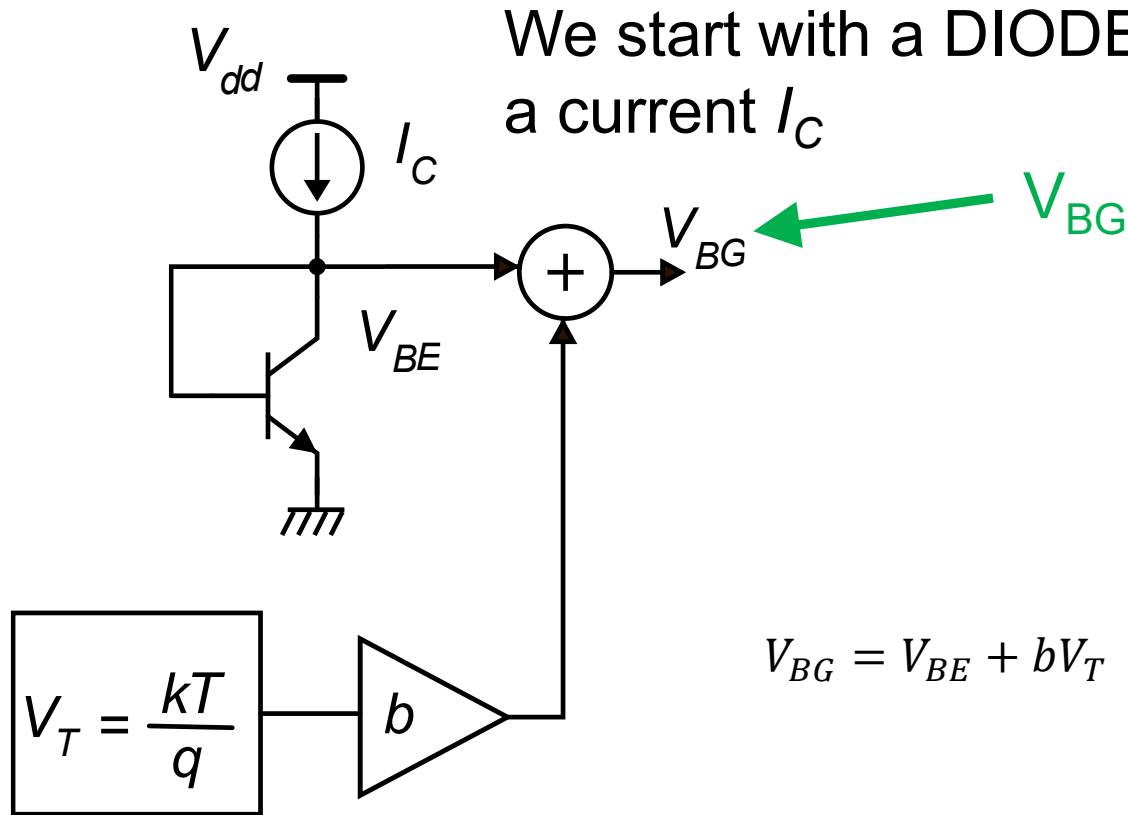
Problems:

- **Require additional process steps**, but only a small number of components are required for each chip (not convenient)
- **Available voltages are $> 3\text{ V}$**
- Temperature stability is poor for $V_Z \neq 5\text{-}6\text{ V}$
- The reference voltage generated by a Zener diode is noisy (very wide band noise)

- Band-gap circuits



Band-gap voltage reference: principle of operation



CTAT: Complementary To Absolute Temperature

PTAT: Proportional To Absolute Temperature

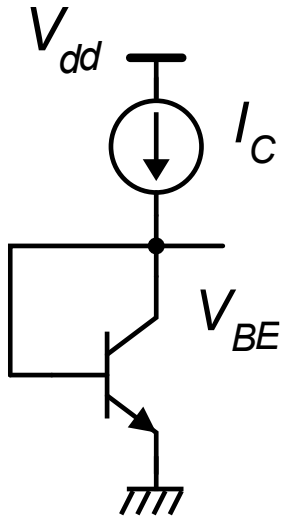
$$\frac{dV_{BE}}{dT} \cong -2\text{mV/K} \dots -3\text{ mV/K}$$

Band-Gap voltage reference: theory

Current of forward-biased diode:

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right) \longrightarrow V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right) = V_T [\ln(I_C) - \ln(I_S)]$$

At first glance one sees that absolute temperature, T , appears in the equation and the quantity V_T , the thermal voltage (kT/q), has a positive temperature coefficient. This would make V_{BE} have a positive temperature coefficient if I_S , the junction saturation current, was constant over temperature. I_S infact has a very strong temperature coefficient as seen in the following equation for I_S .



V_{G0} is the bandgap voltage of silicon "extrapolated to 0°K"

$I_S = BT^\gamma e^{-\frac{V_{G0}}{V_T}} \rightarrow I_S$ increases with temperature rapidly, I_C is constant (current source). To compensate this increase over temperature, V_{BE} decreases (negative temperature coefficient).

Band-gap voltage reference: determination of parameter b

$$V_{BG} = V_{BE} + bV_T$$

$$\frac{dV_{BG}}{dT} = \frac{dV_{BE}}{dT} + b \frac{dV_T}{dT} = 0$$

$$\frac{dV_{BE}}{dT} \cong -2 \text{ mV/K}$$

$$\frac{dV_T}{dT} = \frac{k}{q} \cong 8.56 \times 10^{-5} \text{ V/K}$$

$$V_{BE}(300\text{K}) \approx 0.65\text{V typically}$$

$$\frac{dV_{BE}}{dT} \cong -2 \text{ mV/K}$$

$$V_{BE}(T) \approx V_{BE}(T_0) + \frac{dV_{BE}}{dT}(T - T_0)$$

$$T_0 = 300 \text{ K}$$

We have to determine the value of b , for which:

$$b = -\frac{\frac{dV_{BE}}{dT}}{\frac{dV_T}{dT}} \quad b \cong 23$$

V_{BE}

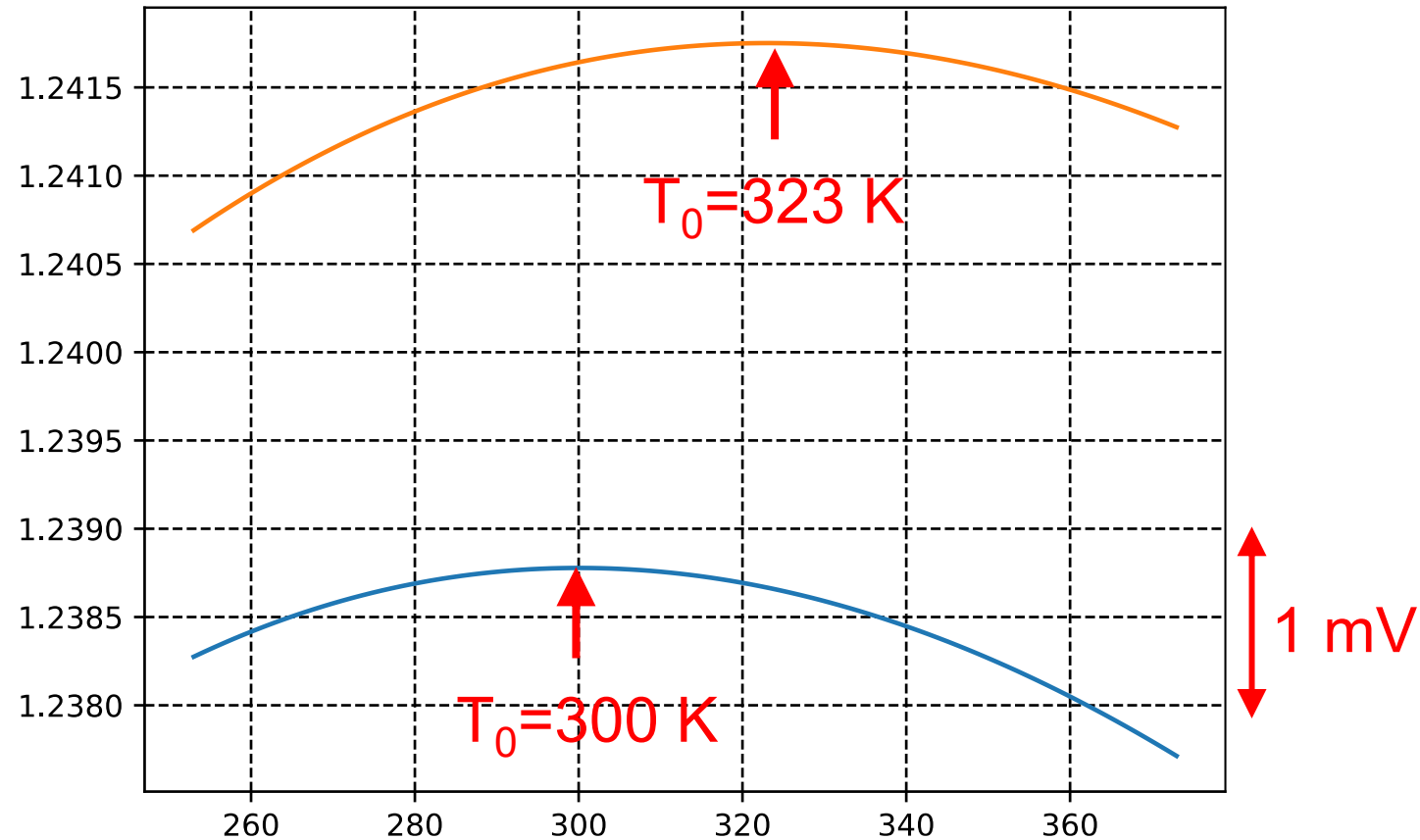
$V_T(290\text{K})$



$$V_{BG} \cong 0.67 + 0.025 \times 23 = 1.245\text{V}$$

$$\frac{dV_{BG}}{dT} = 0$$

Band-gap voltage reference: calculation result



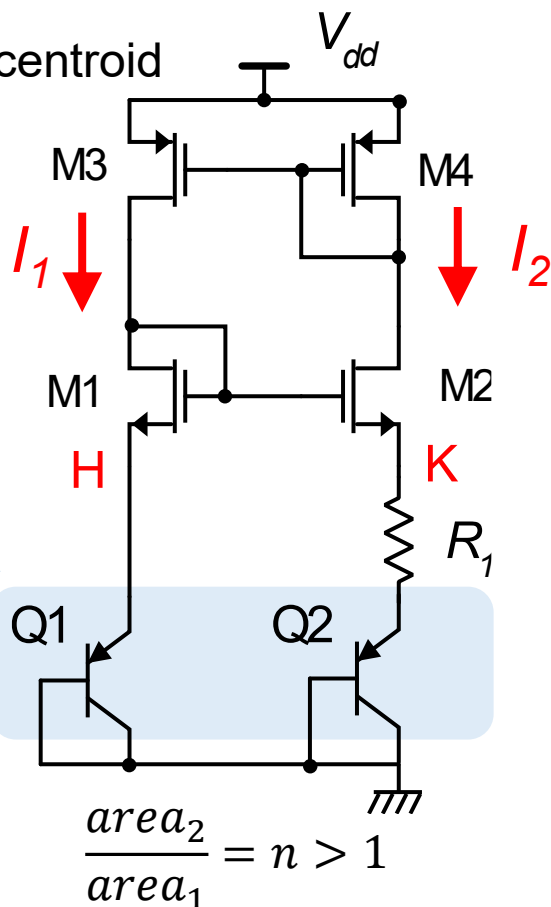
$$V_{BG} = V_{BE} + bV_T$$

Q2	Q2	Q2
Q2	Q1	Q2
Q2	Q2	Q2

Band-Gap voltage: a CMOS compatible Circuit

Part 1: PTAT current generator

Common centroid
layout
For $n=8$



$$M3 = M4 \Rightarrow I_1 = I_2 = I$$

$$M1 = M2$$

neglecting the effects
of V_{DS} on I_D :

$$V_{GS1} = V_{GS2}$$

$$V_H - V_K = (V_{G1} - V_{GS1}) - (V_{G2} - V_{GS2})$$

$$= V_{GS2} - V_{GS1}$$

$$V_H = |V_{BE1}|$$

$$V_K = |V_{BE2}| + R_1 I$$

$$V_H = V_K$$

$$R_1 I = |V_{BE1}| - |V_{BE2}| = V_T \ln \left(\frac{I_{C1} I_{S2}}{I_{S1} I_{C2}} \right)$$

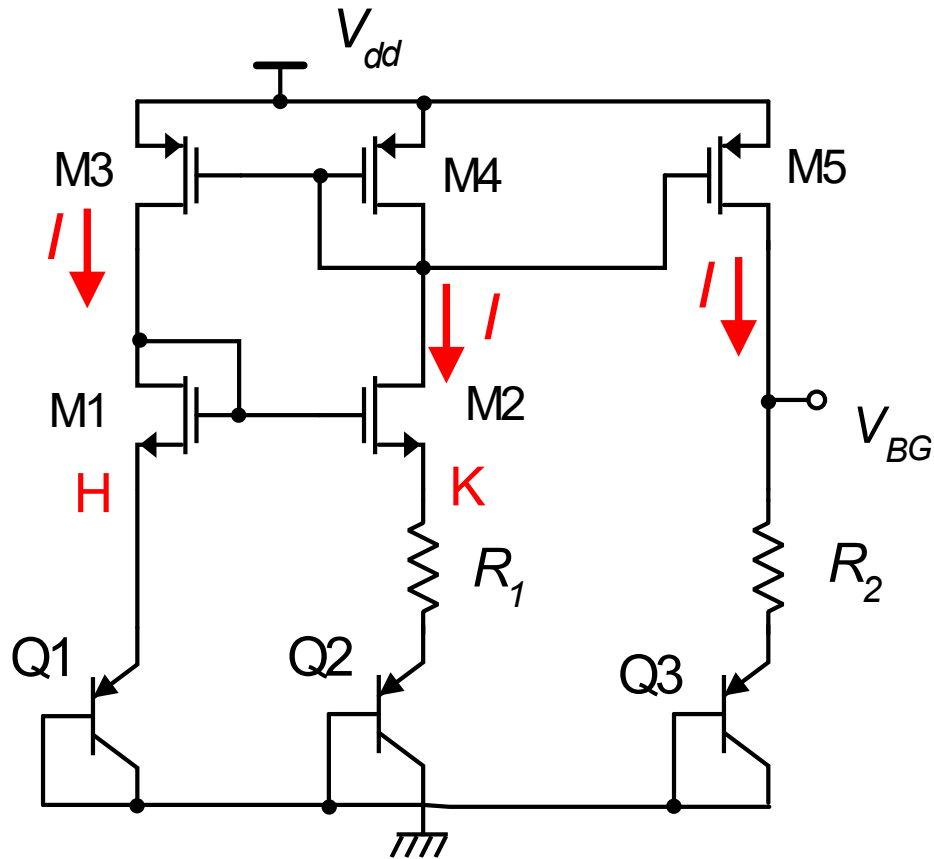
$$I_{S2} = n \cdot I_{S1}$$

$$= V_T \ln(n)$$

$$I = \frac{1}{R_1} \frac{kT}{q} \ln(n)$$

I is proportional to T (PTAT) and
independent of V_{dd} .

Band-Gap voltage: a CMOS compatible Circuit



$$I = \frac{1}{R_1} \frac{kT}{q} \ln(n)$$

$$V_{BG} = |V_{BE3}| + IR_2$$

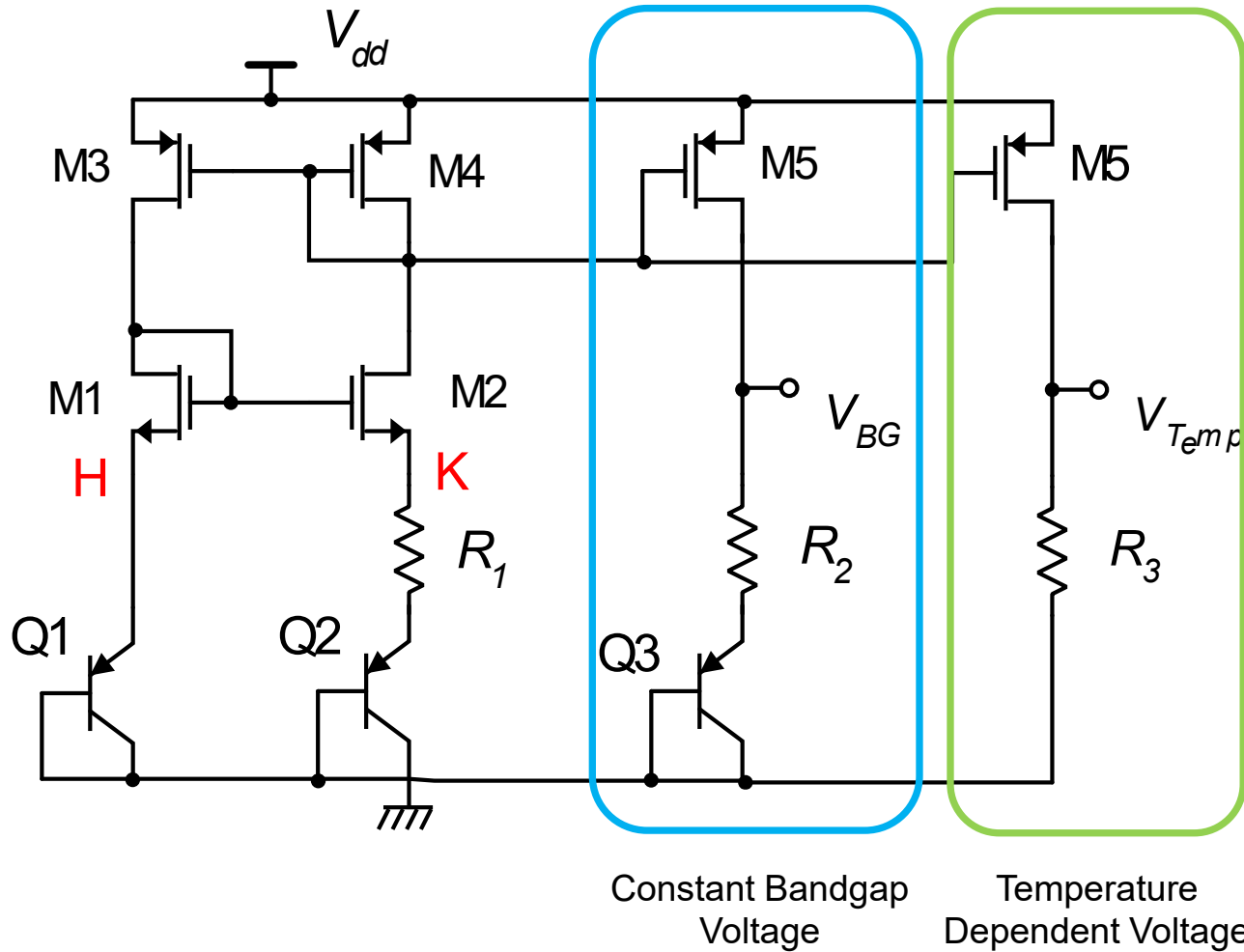
Biased with I
 $\alpha=1$

$$V_{BG} = |V_{BE3}| + \frac{R_2}{R_1} \frac{kT}{q} \ln(n)$$

$$V_{BG} = |V_{BE3}| + V_T \frac{R_2}{R_1} \ln(n)$$

$$V_{BG} = V_{BE} + bV_T$$

Deriving a temperature sensor from the Band-Gap circuit



$$V_{Temp} = \frac{R_3}{R_1} \frac{kT}{q} \ln(n)$$

This output can be used as a temperature sensor!

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.